

Solving Linear Systems of Eqs

Solving the Laplace eq. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ requires solving a linear system of equations, hence we explore methods for solving linear systems.

Jacobi Method

Consider the following system of equations

$$3x - y - z = 1 \quad (1)$$

$$-x + 3y + z = 3 \quad (2)$$

$$2x + y + 4z = 7 \quad (3)$$

To solve this system we solve for x in Eq 1, y in Eq 2, z in eq 3.

$$x = 0x + \frac{1}{3}y + \frac{1}{3}z + \frac{1}{3}$$

$$y = \frac{1}{3}x + 0y - \frac{1}{3}z + 1$$

$$z = -\frac{1}{2}x - \frac{1}{4}y + 0z + \frac{7}{4}$$

Hence,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1 \\ 7/4 \end{bmatrix} + \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/3 & 0 & -1/3 \\ -1/2 & -1/4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

We can write the last equation iteratively like,

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ z_{k+1} \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1 \\ 7/4 \end{bmatrix} + \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/3 & 0 & -1/3 \\ -1/2 & -1/4 & 0 \end{bmatrix} \begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix}$$

Or in vector form like,

$$\vec{r}_{k+1} = \vec{b} + A \vec{r}_k$$

The method works by starting at a initial guess

$\vec{r}_0$, then evolving $\vec{r}_0 \rightarrow \vec{r}_1 \rightarrow \vec{r}_2 \rightarrow \dots \rightarrow \vec{r}_k \rightarrow \vec{r}_{k+1} \rightarrow \dots$

solotion is found when $\vec{r}_{k+1} \approx \vec{r}_k$

We can measure the closeness between $\vec{r}_k$ and $\vec{r}_{k+1}$

by $\|\vec{r}_k - \vec{r}_{k+1}\|$,

Hence we may set a stopping criteria, by setting δ very small,

$$\|\vec{r}_k - \vec{r}_{k+1}\| < \delta$$

Gauss-Seidel Method

This method is an improvement of the Gauss method, it builds the iteration eq. by eq. instead of updating the $\vec{r}_k$ all at once,

Example,

The iteration of the last example can be written as,

$$x_{k+1} = 1/3 + 0 \cdot x_k + (1/3) \cdot y_k + (1/3) z_k \quad (1)$$

$$y_{k+1} = 1 + (1/3) x_{k+1} + (0) y_k + (-1/3) z_k \quad (2)$$

$$z_{k+1} = 7/4 + (-1/2) x_{k+1} + (-1/4) y_{k+1} + (0) z_k \quad (3)$$

We first estimate x_{k+1} from eq. (1), then use the improvement of x_{k+1} on eq(2), and then we use the improvement of x_{k+1} and y_{k+1} to obtain z_{k+1} from eq (3).

References

Segura, Alejandro, Métodos Computacionales, 2021